

Isogeometric Analysis (Basis Functions)

Order Elevation, Order Reduction and Knot Removal

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Element Length

The “element length” is being defined by the control point distance $[1, 2, 3]$. To obtain the effective control point length, we use equally distance control points:

$$\boldsymbol{\xi} = [0, \dots, p], \quad (1)$$

and effective positions

$$\mathbf{d} = \frac{1}{p} \boldsymbol{\xi} (\mathbf{C}^e)^{-1}, \quad (2)$$

where $\mathbf{C}^e$ is the element-based Bézier-extraction operator. From that we can measure the distance of the control points, which can be compared with the actual (physical) element length. This concept [4] is used in the length scale in stabilized finite element formulations.

Locations of the Basis Functions I

With T-splines, where the basis functions are located is not very easy to see. However, using the element-length calculation method, we can obtain one dimensional “positioning” of the basis function.

$$\boldsymbol{\xi} = [0, \dots, p], \quad (3)$$

and effective positions

$$\mathbf{d} = \frac{1}{p} \boldsymbol{\xi} (\mathbf{V})^{-1}. \quad (4)$$

Here $\mathbf{V} \in \mathbb{R}^{(p+1) \times (p+1)}$ is the set of $p + 1$ Bézier-extraction row operators, that have the partition of unity;

$$\sum_{a=0}^p V_{ab} = 1 \quad (5)$$

for $\forall b$. With that if $d_i < d_j$, function i is located at a smaller knot coordinate than that of j . Precisely, the function corresponding to i has smaller knot coordinates of the support range compared with that of j .

Locations of the Basis Functions II

We have another method to order function i and j by just comparing Bézier-extraction row operators. First we normalize the operator as

$$\hat{\mathbf{C}}_i = \frac{1}{|\mathbf{C}_i|} \mathbf{C}_i, \quad (6)$$

where

$$|\mathbf{C}_i| = \sum_{b=0}^p C_{ib}. \quad (7)$$

With that, we can compare $\hat{C}_{ib}$ and $\hat{C}_{jb}$ from $b = 0$ to p . With the smallest b of $\hat{C}_{ib} \neq \hat{C}_{jb}$, if $\hat{C}_{ib} > \hat{C}_{jb}$, function i is located at a smaller knot coordinate than that of j .

Power Basis Functions

The power basis functions are easiest expression for comparing two functions. From Bernstein polynomials, we can obtain power functions:

$$B_a^p(t) = \binom{p}{a} t^a (1-t)^{p-a} \quad (8)$$

$$= \binom{p}{a} t^a \sum_{i=0}^{p-a} \binom{a}{i} (-t)^{p-a-i} \quad (9)$$

$$= \sum_{i=0}^{p-a} \binom{p}{a} \binom{p-a}{i} (-1)^{p-a-i} t^{p-i}. \quad (10)$$

Here,

$$P_{ab}^p = \begin{cases} \binom{p}{b} \binom{b}{a} (-1)^{b-a} & a \leq b \leq p \\ 0 & \text{otherwise} \end{cases}. \quad (11)$$

Here we write $\mathbf{P}^p = [P_{ab}^p]$.

Power Basis Functions — Practice

Write down $\mathbf{P}^2$ and $\mathbf{P}^3$.

Order Elevation I

Similar to knot insertion, order elevation will not change the geometry; it can be a superset of the original function space. This means that $B_a^{p_1}$ can be represented by $B_b^{p_2}$ when $p_1 \leq p_2$:

$$B_a^{p_1} = \sum_{b=0}^{p_2} E_{ab}^{p_1, p_2} B_b^{p_2}, \quad (12)$$

where $\mathbf{E}^{p_1, p_2} = [E_{ab}^{p_1, p_2}] \in \mathbb{R}^{p_1 \times p_2}$ is the order elevation operator. Using power function, we get

$$\mathbf{P}^{p_1} \begin{bmatrix} 1 \\ t \\ \vdots \\ t^{p_1} \end{bmatrix} = \mathbf{E}^{p_1, p_2} \mathbf{P}^{p_2} \begin{bmatrix} 1 \\ t \\ \vdots \\ t^{p_2} \end{bmatrix} \quad (13)$$

Order Elevation II

Since $p_1 \leq p_2$ we can write this as follows:

$$\mathbf{P}^{p_1} \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & & & 0 & 0 & & 0 \\ \vdots & & \ddots & & \vdots & \vdots & & \vdots \\ 0 & & & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ \vdots \\ t^{p_1} \\ \vdots \\ t^{p_2} \end{bmatrix} = \mathbf{E}^{p_1, p_2} \mathbf{P}^{p_2} \begin{bmatrix} 1 \\ t \\ \vdots \\ t^{p_1} \\ \vdots \\ t^{p_2} \end{bmatrix}. \quad (14)$$

$\underbrace{\hspace{10em}}_{p_1+1} \qquad \underbrace{\hspace{10em}}_{p_2-p_1}$

We now write

$$\mathbf{P}^{p_1, p_2} = \left[\begin{array}{c|c} \mathbf{P}^{p_1} & \underbrace{\mathbf{0}}_{\in \mathbb{R}^{(p_1+1) \times (p_2-p_1)}} \end{array} \right] \quad (15)$$

and we get

$$\mathbf{E}^{p_1, p_2} = \mathbf{P}^{p_1, p_2} (\mathbf{P}^{p_2})^{-1}. \quad (16)$$

Determine Control Points

Unlike to knot insertion, the number of control points will change between the conversions. However, it can be obtained very similar to the case in the knot insertion:

$$\sum_{a=0}^{p_2} \sum_{b=0}^{p_2} \bar{\phi}_a \bar{C}_{ab} B_b^{p_2} = \sum_{a=0}^{p_1} \sum_{b=0}^{p_1} \phi_a C_{ab} B_b^{p_1}, \quad (17)$$

and this can be rearranged as

$$\sum_{a=0}^{p_2} \sum_{b=0}^{p_2} \bar{\phi}_a \bar{C}_{ab} B_b^{p_2} = \sum_{a=0}^{p_1} \sum_{b=0}^{p_1} \phi_a C_{ab} \sum_{c=0}^{p_2} E_{bc}^{p_1, p_2} B_c^{p_2}. \quad (18)$$

From that, we get

$$\bar{\phi}_a = \sum_{b=0}^{p_1} \left[\bar{\mathbf{C}}^{-\top} (\mathbf{C} \mathbf{E}^{p_1, p_2})^\top \right]_{ab} \phi_b. \quad (19)$$

We note that $\bar{\mathbf{C}}^{-\top} \in \mathbb{R}^{(p_2+1) \times (p_2+1)}$ and $(\mathbf{C} \mathbf{E}^{p_1, p_2})^\top \in \mathbb{R}^{(p_2+1) \times (p_1+1)}$.

Order Elevation — Practice

Write down $\mathbf{E}^{1,2}$, $\mathbf{E}^{1,3}$, and $\mathbf{E}^{2,3}$. Then, convert a quadratic B-spline curve to the same shape cubic B-spline curve. For each control point converted in all elements are the same. Represent a quadratic B-spline curve by order elevated cubic B-splines, and obtain Bézier control points to draw the curve by using TikZ package in $\text{\LaTeX}$.

Order Reduction — Basis Function Approximation

We now assume $p_1 > p_2$:

$$\tilde{\mathbf{E}}^{p_1, p_2} = (\mathbf{E}^{p_2, p_1})^\top \left((\mathbf{E}^{p_2, p_1}) (\mathbf{E}^{p_2, p_1})^\top \right)^{-1}. \quad (20)$$

Order Reduction — Control Point Approximation

Similar to the order elevation we get:

$$\sum_{a=0}^{p_2} \sum_{b=0}^{p_2} \bar{\phi}_a \bar{C}_{ab} B_b^{p_2} \approx \sum_{a=0}^{p_1} \sum_{b=0}^{p_1} \phi_a C_{ab} B_b^{p_1} \quad (21)$$

$$\approx \sum_{a=0}^{p_1} \sum_{c=0}^{p_1} \phi_a C_{ac} \sum_{b=0}^{p_2} \left[\tilde{\mathbf{E}}^{p_1, p_2} \right]_{cb} B_b^{p_2}, \quad (22)$$

From that, we get

$$\bar{\phi}_a \approx \sum_{b=0}^{p_1} \left[\bar{\mathbf{C}}^{-\top} \left(\mathbf{C} \tilde{\mathbf{E}}^{p_1, p_2} \right)^\top \right]_{ab} \phi_b. \quad (23)$$

We that the control variable obtained from each element may be different.

Knot Removal

In the case of knot removal, again, we cannot represent the same shape. Order reduction may not be able to represent the shape exactly. However, the size of the matrices is the same and we can obtain the control point by the same fashion. The control points we obtain an element may not be the same as the one you obtain from the another element. For example, we get

$$\overline{\phi}_A^e = \sum_{b=0}^p \left[\left(\overline{\mathbf{c}}^e \right)^{-\top} \mathbf{c}^e \right]_{Ab} \phi_b, \quad (24)$$

and

$$\overline{\phi}_A^{e+1} = \sum_{b=0}^p \left[\left(\overline{\mathbf{c}}^{e+1} \right)^{-\top} \mathbf{c}^{e+1} \right]_{(A)b} \phi_b, \quad (25)$$

where A are the global indices, and $\overline{\phi}_A^e$ may not be the same as $\overline{\phi}_A^{e+1}$. In such a case, average over the elements may give a good approximation, and when they are the same, it gives the exact one.

Order Reduction — Practice

Approximate a $p_1 = 3$ Bézier curve by a $p_2 = 2$ Bézier curve. Then, approximate again with two element p_2 B-spline curve.

Knot Removal — Practice

Approximate quadratic B-spline curve by the one with a knot removed. Then, after order elevation, $p = 3$, remove two knots. This resulted in one element less but higher order. You can try with two elements for the first quadratic B-spline curve.

Least-Squares Projection

The least squared projection is a minimization of

$$J = \frac{1}{2} \int_{\Omega} |\phi^h - \phi|^2 d\Omega, \quad (26)$$

where ϕ is the reference solution, and minimized solution is $\delta J = 0$:

$$0 = \int_{\Omega} \delta\phi^h (\phi^h - \phi) d\Omega \quad (27)$$

for $\forall \delta\phi^h$. If we choose the $\delta\phi^h = 1$, we obtain the conservation property of ϕ :

$$\int_{\Omega} \phi^h d\Omega = \int_{\Omega} \phi d\Omega. \quad (28)$$

Least-Squares Projection – Between Two B-Splines Representations I

We consider the following least-squares projection:

$$\int_{\Omega} \delta\phi^C (\phi^C - \phi^F) d\Omega = 0, \quad (29)$$

which means that

$$\int_{\Omega} \delta\phi^C \phi^C d\Omega = \int_{\Omega} \delta\phi^C \phi^F d\Omega \quad (30)$$

$$:= F. \quad (31)$$

We can write

$$F = \sum_a \delta\phi_A^C F_A, \quad (32)$$

$$F_A = \int_{\Omega} N_A^C \phi^F d\Omega, \quad (33)$$

where A is a global index and N_A^C is the corresponding global basis function.

Least-Squares Projection – Between Two B-Splines Representations II

Obtaining the following relationship:

$$N_a^C = \sum_b D_{ab} N_b^F, \quad (34)$$

Eq. (33) can be calculated as follows:

$$F_A = \sum_{d \in E_C^e} \int_{\hat{\Omega}_F^d} N_a^C \sum_b N_b^F \phi_b^F J d\Omega \quad (35)$$

$$= \sum_{d \in E_C^e} \int_{\hat{\Omega}_F^d} \sum_q D_{aq} N_q^F \sum_b N_b^F \phi_b^F J d\Omega \quad (36)$$

$$= \sum_{d \in E_C^e} \sum_b \left(\sum_q \int_{\hat{\Omega}_F^d} D_{aq} N_q^F N_b^F J d\Omega \right) \phi_b^F, \quad (37)$$

where J is the Jacobian in the fine mesh.

Least-Squares Projection — Practice

Test least-squares projection after the knot removal or order reduction to see if there will be a difference.

Reference

- [1] Y. Otaguro, K. Takizawa, and T. E. Tezduyar. “Element Length Calculation in B-Spline meshes for Complex Geometries”. In: *Computational Mechanics* 65 (2020), pp. 1085–1103. DOI: 10.1007/s00466-019-01809-w.
- [2] Y. Ueda et al. “Element-Splitting-Invariant Local-Length-Scale Calculation in B-Spline Meshes for Complex Geometries”. In: *Mathematical Models and Methods in Applied Sciences* 30 (2020), pp. 2139–2174. DOI: 10.1142/S0218202520500402.
- [3] S. Mikawa et al. “Local-length-scale calculation in T-splines meshes for complex geometries”. In: *Mathematical Models and Methods in Applied Sciences* 34.10 (2024), pp. 1933–1970. DOI: 10.1142/S0218202524500404.
- [4] K. Takizawa, T. E. Tezduyar, and Y. Otaguro. “Stabilization and discontinuity-capturing parameters for space–time flow computations with finite element and isogeometric discretizations”. In: *Computational Mechanics* 62 (2018), pp. 1169–1186. DOI: 10.1007/s00466-018-1557-x.